

Does Sentiment Fusion Help?

A Selection-Bias-Free Re-Evaluation of Multimodal Stock Movement Prediction

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Abstract—Multimodal stock-movement prediction—fusing financial-news and social-media sentiment with technical indicators—is widely reported to deliver large risk-adjusted gains, with prior studies reporting Sharpe ratios above two. We show how test-set selection can inflate the apparent performance of a conventional sentiment-fusion pipeline. We build a selection-bias-free evaluation protocol: model-family, horizon, and fusion-weight selection use no test-set information. The protocol combines a chronological train/validation/test split with an embargo and multi-seed reporting with paired significance tests. We apply it to a comprehensive grid of 4,200 task-specific runs (seven model families, eight feature configurations, five U.S. equity instruments, three horizons, five seeds) plus two time-series foundation models (Chronos-2, FinCast). Three findings emerge. *First*, selecting the best model and horizon per stock on the test set inflates the average Sharpe ratio from 1.40 to 2.29, a gap equal to 39% of the test-selected value and comparable to some prior reports. *Second*, under bias-free evaluation a technical-indicator-only baseline is *not* beaten by any sentiment configuration: fixed 70:30 news–social fusion, equal weighting, news-only, social-only, and a learned reliability-weighted fusion all underperform technical-only; the gap is 0.58 Sharpe on average for fixed 70:30 ($p < 10^{-20}$, paired Wilcoxon). *Third*, time-series foundation models do not beat task-specific models and mostly post negative Sharpe ratios. We conclude that, in this setting, the apparent value of the evaluated sentiment fusion is largely explained by optimistic evaluation, and we release a reproducible benchmark to support more careful future studies.

Index Terms—stock movement prediction, multimodal fusion, sentiment analysis, evaluation methodology, selection bias, backtest overfitting, financial machine learning, reproducibility

I. INTRODUCTION

Short-term stock-movement prediction is a long-standing application of data mining, and the dominant recent paradigm is *multimodal*: combine quantitative signals (technical indicators derived from price history) with qualitative signals (sentiment extracted from financial news and social media) [1]–[4]. A steady stream of papers reports that incorporating sentiment produces large, economically meaningful gains, often summarized by trading metrics: long/short strategies driven by news sentiment have reported Sharpe ratios above 2.0 net of transaction costs [5].

These reports sit uneasily beside two well-established facts. First, financial return series have a very low signal-to-noise ratio, and the machine-learning finance literature has repeatedly documented how easily backtests *overfit*: with enough models, horizons, and hyperparameter configurations, a strategy that is indistinguishable from noise can be made to look excellent purely by selection [6]–[10]. Second, when general-purpose machine-learning methods are evaluated against simple baselines under disciplined protocols, the reported advantages frequently shrink or disappear [11].

This paper asks a deliberately narrow, falsifiable question: *under a selection-bias-free evaluation protocol, does multimodal sentiment fusion improve short-term stock-movement prediction relative to a technical-indicator baseline?* We answer it with a comprehensive experimental analysis rather than a new model. We re-implement a conventional multimodal-fusion pipeline—FinBERT sentiment, technical indicators, classical and neural classifiers, a long/short backtest—and evaluate it under a protocol designed to control for selection. Here, “selection-bias-free” refers specifically to the absence of test-set information in model-family, horizon, and fusion-weight selection; it does not rule out validation overfitting (Section VII). The apparent advantage of the evaluated daily FinBERT-Tone fusion does not survive that protocol.

Concretely, we make three contributions.

- **A selection-bias-free evaluation protocol** for multimodal financial prediction (Section III): an explicit train/validation/test chronological split with an embargo gap; *all* model-family and horizon selection locked to the validation window; multi-seed training; and reporting with paired significance tests alongside backtest-overfitting diagnostics [6], [7].
- **A learned reliability-weighted sentiment fusion** that generalizes the fixed 70:30 news–social baseline evaluated here into a data-driven, prior-regularized weight estimated on validation data (Section III-D). With the weight selected using validation only, it does not outperform the technical-only baseline in this setting.
- **A comprehensive re-evaluation** (Section V) across 4,200 task-specific runs and two time-series foundation models, yielding three insights: (i) test-set selection inflates the average Sharpe ratio from 1.40 to 2.29; (ii) no

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evaluated daily FinBERT-Tone sentiment configuration beats a technical-only baseline under bias-free evaluation; (iii) foundation models do not beat task-specific models. We release code and the processed benchmark at GitHub¹.

Our intent is not to claim that sentiment is uninformative in all settings, but to show that a frequently reported positive result does not survive careful evaluation—a comprehensive experimental analysis that yields new insight into the strengths and weaknesses of an existing and widely used method. Fig. 1 summarizes the study: a conventional multimodal pipeline, executed inside a selection-bias-free evaluation protocol.

II. RELATED WORK

Multimodal stock-movement prediction. Combining textual sentiment with market data is now standard. StockNet [1] pairs tweets with prices; MAN-SF [2] adds attentive cross-modal and inter-stock modeling; more recent work fuses news, social media, and engineered indicators [3], [4], [12]. Domain-adapted language models such as FinBERT [13], [14] are the usual sentiment extractors. A recurring design choice is how to combine news and social media. Seetharam and Nyakurukwa [15] find that these sentiment sources are not equivalent, with information flows varying across stocks. We compare a fixed 70:30 news-to-social baseline with equal and learned weights; the 70:30 ratio is a design choice, not an estimate from that study.

Evidence acquisition for multimodal financial reasoning. Recent work shifts from monolithic fusion toward actively acquiring the evidence needed for each reasoning step. Deng et al. propose SEA, a planner-guided multi-agent framework that decomposes financial questions and retrieves task-conditioned evidence from images, text, and external knowledge before answering [16]. Their setting is complementary to ours: SEA studies evidence-grounded multimodal question answering, whereas we test whether pre-aggregated daily sentiment improves out-of-sample movement prediction. It nevertheless motivates event- or query-conditioned text representations as a richer alternative to daily mean pooling.

Time-series foundation models. General-purpose forecasters pre-trained on large, heterogeneous corpora—e.g., Chronos-2 [17] and the finance-specific FinCast [18]—offer zero-shot and fine-tuned prediction and can, in principle, ingest covariates. Whether their broad pre-training transfers to the noisy, non-stationary regime of equity returns is not well characterized.

Evaluation pitfalls in financial ML. The financial-machine-learning literature has long warned that the binding constraint is evaluation, not modeling. López de Prado [8] catalogues leakage and overfitting failure modes; Bailey and López de Prado [6] introduce the Deflated Sharpe Ratio to correct an observed Sharpe ratio for the number of trials conducted; the Probability of Backtest Overfitting [7] quantifies how often an in-sample-best configuration underperforms out of sample; Harvey et al. [10] and Arnott et al. [9] document

multiple-testing inflation across the field. Empirical-asset-pricing studies that adopt disciplined protocols [19] typically report modest, carefully bounded effects. Our contribution is to bring this discipline to bear specifically on the *multimodal-fusion* question, and to quantify how much of the reported advantage it removes.

III. METHODOLOGY AND EVALUATION PROTOCOL

We first state the prediction task, then describe the four elements of the protocol that distinguish bias-free evaluation from the optimistic evaluation common in prior work.

A. Problem Formulation

For a stock s with closing-price series $\{p_t\}$, the binary movement label at horizon h is

$$y_{t+h} = \mathbb{I}\left[\frac{p_{t+h} - p_t}{p_t} > \theta\right], \quad (1)$$

where $\mathbb{I}[\cdot]$ is the indicator function and θ is a per-(stock, horizon) threshold chosen *on the training window* to balance the two classes. A model receives a feature vector $\mathbf{z}_t$ combining technical indicators $\mathbf{X}_t$ and an aggregated daily sentiment score S_t (Section IV), and outputs $\hat{p}(y_{t+h} = 1)$. Predictions are evaluated both as a classification problem (ROC-AUC) and through a long/short trading simulation with non-overlapping h -day positions and 10 bps round-trip transaction costs.

B. Chronological Split with Embargo

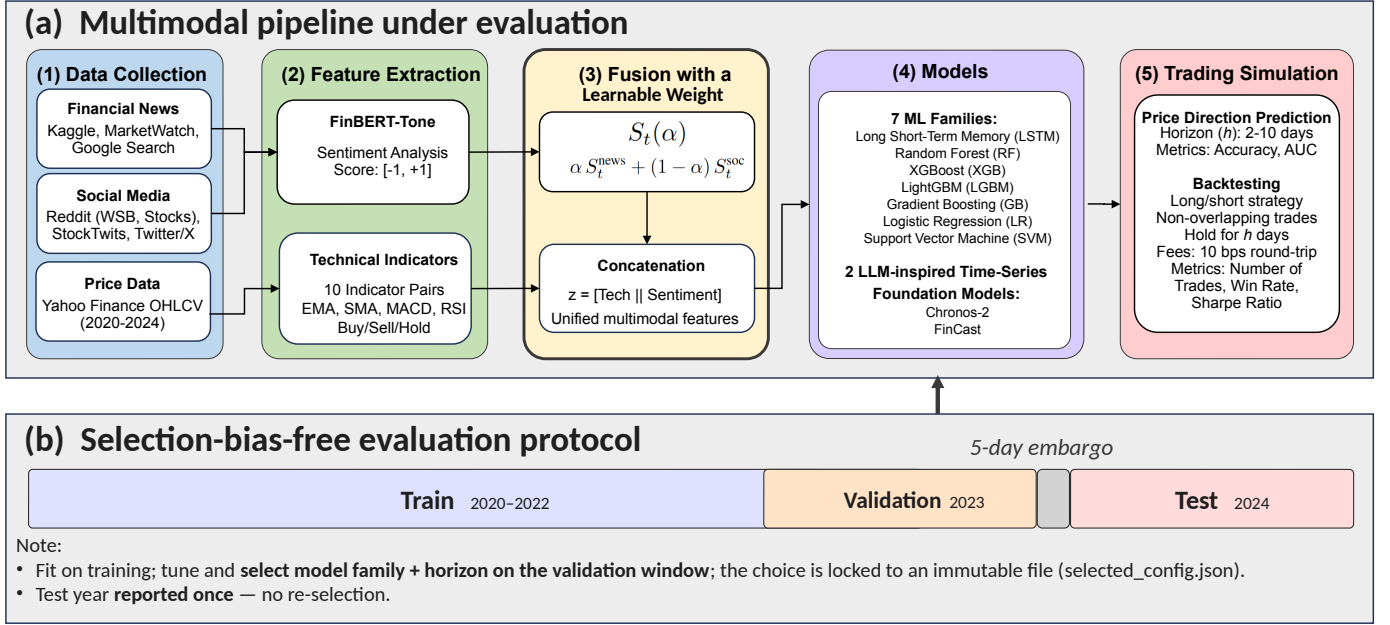
All data span January 2020 to December 2024. We use a strict calendar split: **training** 2020–2022, **validation** 2023, **test** 2024. The last h rows of the validation window are dropped so that validation labels, which look h days ahead, cannot reach into the test year, and a further five-trading-day *embargo* is removed from the start of the test window [8]. This replaces the train/test-only split of prior pipelines, which leaves no clean data on which to make modeling choices.

C. Validation-Locked Model and Horizon Selection

This is the single most consequential element. Prior multimodal pipelines typically report, for each stock, the *best* model family and prediction horizon—but the “best” is identified on the test year. Even our grid of seven model families and three candidate horizons creates 21 trials per stock; the maximum over noisy trials is an upward-biased estimate of true performance [6], [10].

We instead compute, for every (stock, ablation, model, horizon) cell, the *validation* ROC-AUC and validation Sharpe ratio, and write the selection—which model and which horizon—to an immutable configuration file keyed only by validation scores. Every number reported on the 2024 test year is then read back through that file. No test-set quantity is ever used to make a modeling choice. Section V-A quantifies how much this single change removes.

¹<https://github.com/inflaton/Multimodal-Stock-Movement-Prediction>



Grid: 8 feature sets $\times$ 7 models $\times$ 5 stocks $\times$ 3 horizons $\times$ 5 seeds = **4,200 task-specific runs** (plus 2 time-series foundation models). Reported as multi-seed mean $\pm$ std with paired Wilcoxon signed-rank tests.

Fig. 1. Overview of the study. (a) A conventional multimodal stock-movement pipeline: news, social-media, and price data are turned into FinBERT-Tone sentiment and technical-indicator features, fused, fed to seven model families, and evaluated by a long/short backtest. (b) The selection-bias-free evaluation protocol that is the methodological contribution: a chronological train/validation/test split with a five-day embargo; *all* model-family and horizon selection locked to the 2023 validation window; and a 4,200-run multi-seed grid reported with paired significance tests. Every grid cell is one execution of pipeline (a).

D. Learned Reliability-Weighted Fusion

Our fixed baseline aggregates daily news sentiment S_t^{news} and social sentiment S_t^{soc} with 70:30 weights, a news-weighted design choice:

$$S_t = 0.7 S_t^{\text{news}} + 0.3 S_t^{\text{soc}}, \quad (2)$$

We generalize Equation 2 by introducing a learnable weight:

$$S_t(\alpha) = \alpha S_t^{\text{news}} + (1 - \alpha) S_t^{\text{soc}} \quad (3)$$

where $\alpha \in [0, 1]$ is a learnable weight estimated by maximizing the validation ROC-AUC with a quadratic penalty pulling α toward the fixed baseline value of 0.7:

$$\alpha^* = \arg \max_{\alpha} \text{AUC}_{\text{val}}(\alpha) - \lambda (\alpha - 0.7)^2. \quad (4)$$

The fixed 70:30 rule is recovered as $\lambda \rightarrow \infty$. We fit α both per stock and globally (one shared α). This makes the weighting a data-driven choice that the evaluation can test rather than assume.

E. Multi-Seed Reporting and Statistics

Every stochastic configuration is trained with five random seeds; we report the mean and standard deviation across seeds. Because the central question is a *paired* comparison—does configuration *A* beat configuration *B* on the same stock, model, horizon, and seed?—we use the Wilcoxon signed-rank test over all matched cells, which makes no normality assumption. We additionally adopt the Deflated Sharpe Ratio

[6] and the Probability of Backtest Overfitting [7] as backtest-overfitting diagnostics; their implementations are included in the released code. Trading metrics reported are the Sharpe ratio [20], win rate, cumulative return over the test year, and maximum drawdown (MDD).

IV. EXPERIMENTAL SETUP

A. Data

We study four large-cap U.S. stocks (AAPL, META, NVDA, and TSLA) and the SPY equity ETF, spanning different market exposures. Daily OHLCV prices are from Yahoo! Finance [21]. Financial-news and social-media text (2020–2024) is scored with FinBERT-Tone [14] to produce a per-document sentiment in $[-1, 1]$, then averaged per trading day into S_t^{news} and S_t^{soc} . Table I summarizes coverage. Days lacking text of a given type are imputed as neutral (0); such days are rare except for SPY news, which we revisit in Section VII. A first warning sign is visible already here: the correlation between daily news sentiment and next-day return is tiny and sign-inconsistent across stocks (between -0.05 and $+0.06$).

B. Feature Configurations (Ablations)

The technical branch follows prior work [4]: ten indicator-pair signals per stock (moving-average, MACD, RSI, and Bollinger-band crossovers), each encoded as Buy/Sell/Hold $\in \{+1, -1, 0\}$ and selected on the training window only. We evaluate eight feature configurations: **technical_only**

TABLE I
DATASET COVERAGE, 2020–2024 (1,257 TRADING DAYS PER STOCK).
“COV.” IS THE FRACTION OF TRADING DAYS WITH AT LEAST ONE
DOCUMENT OF THAT TYPE.

Stock	News cov.	Social cov.	Mean S^{news}	Mean S^{soc}
AAPL	0.993	0.996	0.456	0.690
META	0.936	0.887	0.521	0.803
NVDA	0.937	0.994	0.455	0.665
SPY	0.835	0.947	0.371	0.767
TSLA	1.000	0.991	0.473	0.752

(indicators, no sentiment); **full_70_30** (indicators + fixed 70:30 fusion); **equal_50_50** (indicators + 50:50 fusion); **news_only** and **social_only** (indicators + single-source sentiment); **sentiment_only** (fused sentiment, no indicators); **learned_alpha_per_stock** and **learned_alpha_global** (indicators + the learned fusion of Section III-D).

C. Models

Task-specific. Seven families: Logistic Regression (LR), SVM, Random Forest [22], Gradient Boosting, XGBoost [23], LightGBM [24], and a two-layer LSTM [25]. Each is tuned with 30 iterations of Bayesian optimization [26] on the validation window. **Foundation.** Chronos-2 [17] in four configurations (zero-shot and fine-tuned, each with and without sentiment/indicator covariates) and FinCast [18], whose released artifact supports only price-only inputs.

D. Grid and Implementation

The task-specific grid is 7 models $\times$ 8 ablations $\times$ 3 horizons $\{2, 5, 10\} \times 5$ seeds $\times 5$ stocks = 4,200 runs; foundation models add 75 runs. Backtesting uses a long/short strategy, non-overlapping h -day positions, and 10 bps round-trip costs.

V. RESULTS AND ANALYSIS

A. Selection Bias Inflates the Headline

We first isolate the effect of test-set selection. For the fixed-70:30 baseline pipeline, we select the best (model, horizon) per stock two ways: *test-peek*, maximizing the 2024 test Sharpe ratio (the optimistic procedure implicit in reporting per-stock maxima), and *validation-locked*, maximizing the 2023 validation Sharpe ratio and then reading the test number back. Table II shows the result. Test-peek selection yields a 2.29 average Sharpe ratio—squarely in the range of some prior reports—while the identical pipeline, selected using validation only, yields 1.40. Roughly 0.9 units of Sharpe ratio, about 39% of the test-peek value, is attributable to the selection rule alone. This is a controlled reconstruction of the mechanism, not a direct reproduction of any specific prior paper. Discrimination is barely affected (AUC 0.560 vs. 0.565): the bias lives almost entirely in the trading metric, exactly where overfitting concentrates [6].

TABLE II
SELECTION BIAS: BEST (MODEL, HORIZON) PER STOCK FOR THE
FIXED-70:30 PIPELINE, SELECTED ON THE TEST SET VS. THE VALIDATION
SET. TEST-YEAR SHARPE RATIO AND ROC-AUC.

Stock	Test-peek selection		Validation-locked	
	Sharpe	AUC	Sharpe	AUC
AAPL	0.55	0.529	−0.45	0.533
META	1.89	0.586	1.89	0.586
NVDA	4.14	0.599	4.14	0.599
SPY	2.33	0.571	1.11	0.607
TSLA	2.57	0.514	0.30	0.502
Average	2.29	0.560	1.40	0.565

B. Does Sentiment Fusion Help in This Setting?

The central experiment compares the eight feature configurations under bias-free evaluation. Table III reports the mean over the entire grid—all seven model families, three horizons, five stocks, five seeds (525 runs per configuration)—which removes any dependence on a hand-picked model or horizon.

A note on aggregation, since two views of the data appear in this paper. Table II reports the *best configuration per stock*—one model and one horizon chosen for each stock—which is why its validation-locked figure for fixed 70:30 is 1.40. Table III instead reports the *grid mean*: the average over every model and horizon with no selection at all, 0.23 for the same configuration. These are not in conflict—one is a favourable per-stock pick, the other an unselected average—and the grid mean summarizes performance across the evaluated grid without selecting a model or horizon. Table IV reports both views for all eight configurations: **technical_only** ranks first under *both* the grid mean and the per-stock best-configuration view, while the ordering of the sentiment configurations is itself unstable between the two—per-stock selection inflates every configuration and reshuffles their ranks, whereas the grid mean exposes the gaps. The evaluated sentiment configurations underperform technical-only under both aggregation schemes.

The pattern is consistent across metrics. **technical_only** attains the highest Sharpe ratio (0.81), the highest AUC (0.542), the highest win rate, and the shallowest drawdown. *Every* configuration that adds sentiment does worse, and the paired Wilcoxon test rejects equality at $p < 2 \times 10^{-7}$ for all seven comparisons, with technical-only winning 57–70% of the 525 matched cells. Adding fixed 70:30 fusion costs 0.58 Sharpe units on average; replacing indicators entirely with sentiment (**sentiment_only**) costs 1.11 and drives both AUC and win rate below chance. The controlled single-model view (LightGBM at $h = 5$) tells the same story—technical-only 0.95 Sharpe vs. 0.23 for full_70_30—so the finding is not an artifact of pooling.

These results do not support claims that fusion is “essential” or that sentiment “drives profitability” in this setting. Under the evaluated protocol, adding the daily mean-pooled FinBERT-Tone sentiment representation does not improve average performance and can degrade it; this finding does not establish

TABLE III

FEATURE-CONFIGURATION COMPARISON UNDER BIAS-FREE EVALUATION. EACH ROW IS THE MEAN OVER 525 RUNS (7 MODELS $\times$ 3 HORIZONS $\times$ 5 STOCKS $\times$ 5 SEEDS); SHARPE SHOWN AS MEAN $\pm$ STD. Δ SHARPE AND p ARE THE PAIRED WILCOXON COMPARISON OF **TECHNICAL_ONLY** MINUS THAT ROW, OVER ALL 525 MATCHED CELLS. POSITIVE Δ MEANS TECHNICAL-ONLY WINS.

Configuration	Test AUC	Test Sharpe	Win rate	Cum. return	MDD	Δ Sharpe vs. tech.	p (Wilcoxon)
technical_only	0.542	0.81 $\pm$ 1.61	0.542	0.73	−0.265	—	—
equal_50_50	0.536	0.46 $\pm$ 1.45	0.506	0.45	−0.275	+0.35	1.1×10^{-7}
learned_alpha_per_stock	0.532	0.39 $\pm$ 1.40	0.508	0.38	−0.272	+0.42	3.6×10^{-10}
news_only	0.541	0.24 $\pm$ 1.39	0.504	0.33	−0.283	+0.57	6.4×10^{-20}
full_70_30	0.532	0.23 $\pm$ 1.46	0.496	0.37	−0.295	+0.58	2.4×10^{-21}
learned_alpha_global	0.530	0.21 $\pm$ 1.51	0.495	0.37	−0.295	+0.61	1.9×10^{-15}
social_only	0.515	−0.05 $\pm$ 1.33	0.482	0.09	−0.322	+0.86	4.5×10^{-30}
sentiment_only	0.499	−0.30 $\pm$ 1.18	0.469	−0.06	−0.346	+1.11	4.0×10^{-31}

TABLE IV

THE HEADLINE RANKING IS INVARIANT TO AGGREGATION. *Grid-mean Sharpe*: MEAN TEST SHARPE OVER ALL 525 RUNS OF A CONFIGURATION (THE TEST SHARPE COLUMN OF TABLE III). *Best-per-stock Sharpe*: FOR EACH STOCK THE (MODEL, HORIZON) WITH THE HIGHEST *validation* SHARPE IS SELECTED AND ITS TEST SHARPE RECORDED; THE FIVE ARE AVERAGED. **TECHNICAL_ONLY** RANKS FIRST UNDER BOTH; PER-STOCK SELECTION INFLATES EVERY CONFIGURATION AND RESHUFFLES THE REST.

Configuration	Grid-mean Sharpe	Best-per-stock Sharpe
technical_only	0.81	2.26
equal_50_50	0.46	1.13
learned_alpha_per_stock	0.39	2.06
news_only	0.24	1.20
full_70_30	0.23	1.40
learned_alpha_global	0.21	2.11
social_only	−0.05	1.20
sentiment_only	−0.30	0.16

that other textual representations are uninformative.

C. The Learned Fusion Does Not Rescue It

The learned reliability weight (Section III-D) was our intended methodological improvement. It behaves sensibly—Table V and Fig. 4 show α^* landing near or above the baseline value of 0.7 for four of five stocks—but it does not change the conclusion. On validation AUC it beats the fixed 70:30 rule by a small margin for four stocks ($\Delta < 0.006$); only META shows a real gain (+0.047). On the test year, **learned_alpha_per_stock** (0.39 Sharpe) and **learned_alpha_global** (0.21) still lose to technical-only by 0.42 and 0.61 Sharpe units respectively (Table III, $p < 10^{-9}$). Thus, optimizing this single convex weight on validation data does not make the evaluated daily sentiment representation outperform technical indicators alone.

D. Model Families: Logistic Regression Leads on Average

Table VI breaks the task-specific grid down by model family (each row averaged over all eight feature configurations, three horizons, five stocks, and five seeds). Logistic Regression, a regularized linear classifier, has the highest mean AUC (0.547), Sharpe ratio (0.90), and win rate (0.542) in this

TABLE V

LEARNED NEWS WEIGHT α^* VS. THE FIXED 0.7 BASELINE, AND THE VALIDATION-AUC GAIN. ESTIMATED ON 2023 VALIDATION DATA WITH PRIOR STRENGTH $\lambda = 0.1$.

Stock	α^*	Val AUC (learned)	Δ vs. $\alpha=0.7$
AAPL	0.72	0.563	+0.001
META	0.84	0.618	+0.047
NVDA	0.82	0.656	+0.005
SPY	0.80	0.577	+0.004
TSLA	0.54	0.584	+0.004
Global	0.80	0.598	+0.011

TABLE VI

TASK-SPECIFIC MODEL FAMILIES, EACH AVERAGED OVER THE FULL FEATURE $\times$ HORIZON $\times$ STOCK $\times$ SEED GRID (600 RUNS PER FAMILY).

Model family	Test AUC	Test Sharpe	Win rate
Logistic Regression	0.547	0.90	0.542
Random Forest	0.539	0.26	0.494
XGBoost	0.533	0.36	0.503
SVM	0.531	−0.06	0.478
Gradient Boosting	0.528	0.10	0.487
LightGBM	0.519	0.24	0.501
LSTM	0.499	−0.04	0.497

comparison. More expressive models do not show a consistent advantage: the LSTM has near-chance AUC (0.499), while both the LSTM and SVM post slightly negative average Sharpe ratios (−0.04 and −0.06, respectively). These aggregate results favor Logistic Regression for the evaluated features and tuning budgets; they do not establish a general ranking by model capacity or regularization strength.

E. Foundation Models Do Not Beat Task-Specific Models

Table VII compares the five foundation-model configurations against the task-specific family with the highest mean test Sharpe. Every foundation configuration posts a near-chance AUC (0.47–0.50) and, in aggregate, a negative Sharpe ratio. Fine-tuning improves mean AUC in both input settings, but its effect on Sharpe is mixed: without covariates, Sharpe falls from −0.38 to −0.54; with covariates, it rises from

TABLE VII

FOUNDATION MODELS VS. THE TASK-SPECIFIC FAMILY WITH THE HIGHEST MEAN TEST SHARPE. FOUNDATION ROWS ARE MEANS OVER 15 RUNS (5 STOCKS $\times$ 3 HORIZONS, SINGLE SEED); THE TASK-SPECIFIC ROW IS THE MEAN OVER ITS FULL GRID.

Method	Test AUC	Test Sharpe	Win rate
Chronos-2 zero-shot	0.472	−0.38	0.454
Chronos-2 zero-shot + cov.	0.482	−0.61	0.436
Chronos-2 fine-tuned	0.497	−0.54	0.450
Chronos-2 fine-tuned + cov.	0.484	−0.02	0.470
FinCast (price-only) [†]	0.498	−0.70	0.425
Task-specific (LR)	0.547	0.90	0.542

[†]The released FinCast artifact does not accept covariates; this is an artifact limitation, not a property of the model class.

−0.61 to −0.02. The fine-tuned-with-covariates variant is the only one to approach break-even. But none matches the task-specific models: Logistic Regression, the best family on the test grid, reaches 0.90 average Sharpe and 0.547 AUC, and even the mean over all seven families (0.25 Sharpe, 0.528 AUC) exceeds every foundation configuration. The often-quoted claim that “lightweight task-specific models beat foundation models” thus holds in our data—but it is a weak consolation, because within the task-specific grid, adding the evaluated sentiment features does not improve on technical indicators alone (Table III).

F. Where Any Signal Lives: Horizon and Discrimination

Two secondary patterns help interpret the main result. First, mean Sharpe is highest at the shortest horizon: averaged over the grid, $h = 2$ gives 0.51 Sharpe and 0.527 AUC, decaying to 0.04 Sharpe at $h = 10$ (Fig. 2); mean AUC instead peaks at $h = 5$ (0.536). Second, win rate and Sharpe ratio are strongly positively correlated across the grid (Pearson $r = 0.830$; Fig. 3), but some configurations clear a 50% win rate yet still have negative Sharpe, consistent with a few large adverse moves dominating returns. The horizon pattern is consistent with a low signal-to-noise ratio that decays quickly with horizon.

VI. DISCUSSION

Why does fusion fail here? One possible explanation is limited predictive information in the evaluated representation: daily FinBERT-Tone sentiment correlates with next-day returns at $|\rho| \leq 0.06$, with inconsistent sign across stocks. These small linear correlations do not rule out nonlinear or longer-horizon relationships, but the ablations show no average improvement from adding this representation (Table III). Aggregating thousands of documents into one daily scalar discards most of the textual content; the scalar that remains is dominated by a persistently positive tone (Table I) that varies little with the next move.

Why is this not visible in prior work? Two reasons. First, test-set selection (Table II) manufactures roughly 0.9 Sharpe units for the fixed-70:30 configuration; a pipeline that quotes the per-stock maximum will look strong whether or

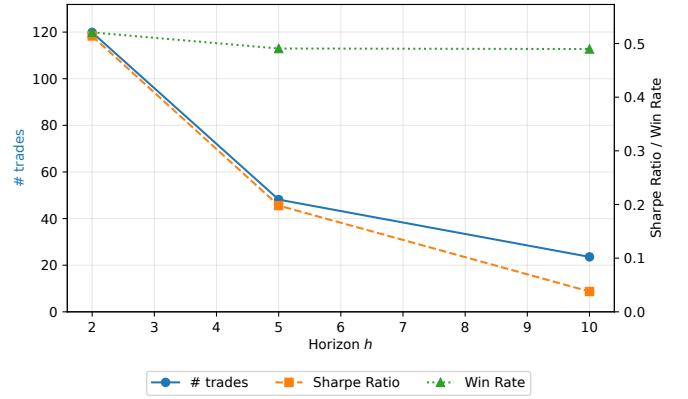


Fig. 2. Mean number of trades, Sharpe ratio, and win rate by horizon over the full grid. Mean Sharpe is highest at the shortest horizon and declines as h increases.

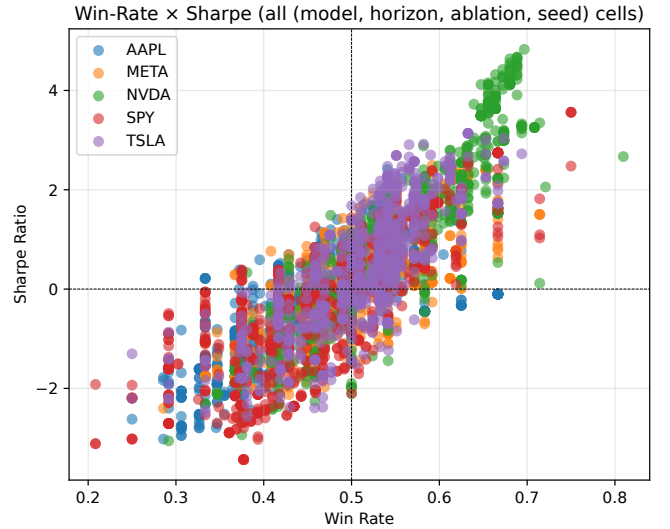


Fig. 3. Win rate vs. Sharpe ratio across all task-specific runs. Despite a strong positive correlation ($r = 0.830$), a win rate above 0.5 does not guarantee a positive Sharpe ratio once transaction costs and return asymmetry are accounted for.

not sentiment helps. Second, isolated-modality ablations are typically read on the trading metric alone, which is dominated by variance at small trade counts; a careful paired, multi-seed comparison—absent from most prior pipelines—tells a different story.

Implications. The contribution of this paper is therefore methodological and cautionary. The selection-bias-free protocol of Section III is inexpensive, transferable, and, in our case, consequential: it converts an apparently strong multimodal result—an average Sharpe ratio above two under test-set selection (Table II)—into a validation-locked finding that, here, daily FinBERT-Tone sentiment fusion does not help. We do not claim sentiment is universally useless—higher-frequency text, event-level rather than daily aggregation, or richer representations than a scalar may yet carry signal—but we do

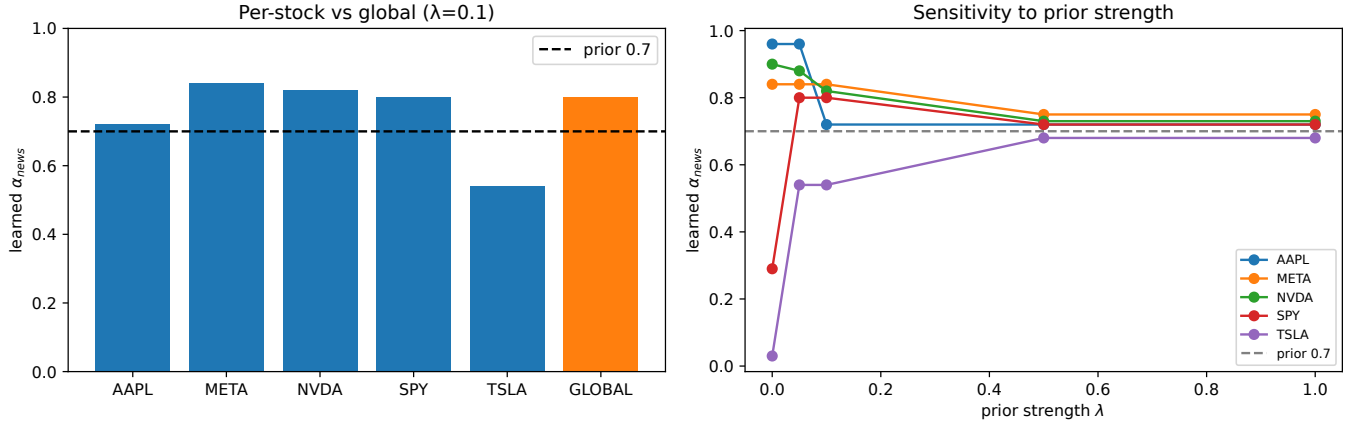


Fig. 4. Learned news weight α^* per stock (and global), and its sensitivity to the prior strength λ . At $\lambda = 0.1$, per-stock weights range from 0.54 to 0.84; stronger penalties pull them toward 0.7. The configurations built on the default weights still lose to technical-only (Table III).

claim that any such result must be demonstrated under an evaluation protocol that controls for selection. We recommend that multimodal-finance studies (i) report a validation-locked technical-only baseline, (ii) report paired, multi-seed significance tests rather than per-stock maxima, and (iii) accompany Sharpe ratios with the Deflated Sharpe Ratio for the number of trials conducted [6], [7].

VII. LIMITATIONS

Our study is deliberately a focused re-evaluation, and its scope bounds its claims. (i) **Universe:** four large-cap U.S. stocks, SPY, and a single test year (2024); the negative result may not transfer to other regimes, small caps, or other markets, though the selection-bias mechanism is general. (ii) **Sentiment representation:** one daily FinBERT-Tone scalar per source; document embeddings, event tags, intraday text, or step-wise evidence acquisition [16] could preserve signal that mean pooling discards. (iii) **Data quality:** SPY news coverage in the test year is only 41%, so SPY’s sentiment ablations are partly imputed. Excluding SPY from the saved test results leaves 420 runs per feature configuration. Mean Sharpe is 1.05 for technical-only, 0.39 for fixed 70:30 fusion (gap 0.65), and 0.58 for the strongest sentiment configuration (per-stock learned fusion). Technical-only also retains the highest mean AUC (0.543 vs. 0.540 for the strongest sentiment configuration by AUC, news-only). This check excludes SPY from evaluation only; fitted models and fusion weights are unchanged. (iv) **Foundation-model comparison:** these models were run single-seed under compute limits, and the released FinCast artifact accepts only prices, so this comparison is directional rather than conclusive. (v) **Fusion class:** our learned method is a one-parameter convex weight; its failure does not rule out richer cross-modal architectures. (vi) **Prior-work comparison:** the 2.29 test-peek result reconstructs a selection mechanism in our pipeline rather than rerunning another paper’s code; it demonstrates the size of the bias but does not attribute any specific prior result to that mechanism. (vii) **Validation**

reuse: the same 2023 window is used for hyperparameter tuning, model-family and horizon selection, and fusion-weight estimation. Repeated optimization can overfit this validation window even without test-set information. Nested temporal validation or additional rolling validation windows would better assess sensitivity to these choices. (viii) **Test dependence:** the 525 matched cells are not independent, because the five seeds share each stock, model, and horizon, so cell-level p -values overstate the evidence. Averaging over seeds first (105 cells), technical-only still beats every sentiment configuration at $p < 0.01$.

VIII. CONCLUSION

We re-evaluated the reported benefits of multimodal sentiment fusion for short-term stock-movement prediction under a selection-bias-free protocol: an explicit chronological split with embargo, validation-locked model and horizon selection, multi-seed training, and paired significance testing. Across 4,200 task-specific runs and two time-series foundation models, the apparent advantage of the evaluated fusion did not survive. Test-set selection alone inflates the average Sharpe ratio from 1.40 to 2.29; under validation-locked evaluation in this setting, no evaluated daily FinBERT-Tone sentiment configuration—including a learned reliability-weighted fusion that generalizes the fixed 70:30 baseline—beats a technical-indicator-only baseline, and foundation models do not beat task-specific models. The practical recommendation is not to abandon textual data, but to abandon optimistic evaluation: the value of a multimodal pipeline must be demonstrated against a validation-locked baseline with paired, multi-seed statistics. We release the protocol, code, and processed benchmark to make that the path of least resistance for future work.

REPRODUCIBILITY

All code, configuration files, processed per-stock features, and the full results table are released at GitHub, together with the immutable validation-locked selection file that all reported test numbers are read through.

APPENDIX A PER-SPLIT DATASET STATISTICS

Table VIII reports text coverage and mean news sentiment separately for the training, validation, and test windows. The notable irregularity is SPY news coverage in the test year (0.41), discussed in Section VII.

TABLE VIII

TEXT COVERAGE BY SPLIT. TRAIN: 2020–2022; VAL: 2023; TEST: 2024.

Stock	Split	News cov.	Social cov.	Mean S^{news}
AAPL	train	0.988	1.000	0.489
	val	1.000	0.992	0.466
	test	1.000	0.988	0.350
META	train	0.938	0.897	0.542
	val	0.968	0.836	0.447
	test	0.900	0.908	0.537
NVDA	train	0.896	1.000	0.481
	val	1.000	0.972	0.478
	test	1.000	1.000	0.359
SPY	train	0.927	0.918	0.393
	val	0.984	0.984	0.261
	test	0.406	1.000	0.486
TSLA	train	1.000	1.000	0.512
	val	1.000	0.968	0.435
	test	1.000	0.988	0.392

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